Kehrlage: $\omega_p > \omega_s \Rightarrow \omega_i = \omega_p - \omega_s \qquad \omega_s = \omega_p - \omega_i$

$$-\omega_i = -\omega_p + \omega_s \qquad -\omega_s = -\omega_p + \omega_i$$

$$\Delta i(t) = \frac{1}{2}\left[I_s e^{j\omega_s t} + I_s^* e^{-j\omega_s t} + I_i e^{j\omega_i t} + I_i^* e^{-j\omega_i t} \right]$$

$$= \sum_{n=-\infty}^{\infty} G_n e^{jn\omega_p t} \cdot \frac{1}{2}\left[U_s e^{j\omega_s t} + U_s^* e^{-j\omega_s t} + U_i e^{j\omega_i t} + U_i^* e^{-j\omega_i t} \right]$$

n=0
n=1
n=-1

$$= \frac{1}{2}\Big[G_0 U_s e^{j\omega_s t} + G_0 U_s^* e^{-j\omega_s t} + G_0 U_i e^{j\omega_i t} + G_0 U_i^* e^{-j\omega_i t} + G_1 U_s^* e^{j\underbrace{(\omega_p-\omega_s)}_{\omega_i} t} + G_1 U_i^* e^{j\underbrace{(\omega_p-\omega_i)}_{\omega_s} t}$$

$$+ G_{-1} U_s e^{j\underbrace{(-\omega_p+\omega_s)}_{-\omega_i}} + G_{-1} U_i e^{j\underbrace{(-\omega_p+\omega_i)}_{-\omega_s}} \Big]$$

nach Frequenzkomponenten:

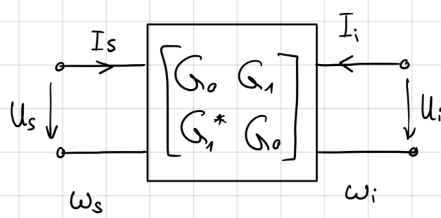
$$\left.\begin{aligned} \cancel{\tfrac{1}{2}} I_s \cancel{e^{j\omega_s t}} &= \cancel{\tfrac{1}{2}}\left[G_0 U_s \cancel{e^{j\omega_s t}} + G_1 U_i^* \cancel{e^{j\omega_s t}} \right] \\ \cancel{\tfrac{1}{2}} I_i \cancel{e^{j\omega_i t}} &= \cancel{\tfrac{1}{2}}\left[G_0 U_i \cancel{e^{j\omega_i t}} + G_1 U_s^* \cancel{e^{j\omega_i t}} \right] \end{aligned}\right\} \begin{bmatrix} I_s \\ I_i^* \end{bmatrix} = \begin{bmatrix} G_0 & G_1 \\ G_1^* & G_0 \end{bmatrix} \begin{bmatrix} U_s \\ U_i^* \end{bmatrix}$$

Kehrlage mit $\omega_s < \omega_p$

$*$

$$I_i^* = G_0 U_i^* + G_1^* U_s$$

$\Rightarrow$



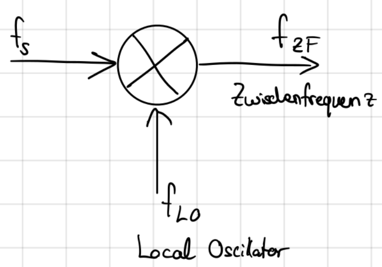
$\omega_s > \omega_p$ Gleichlage

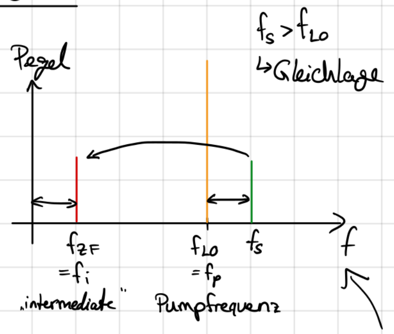
$$\begin{bmatrix} I_s \\ I_i \end{bmatrix} = \begin{bmatrix} G_0 & G_1 \\ G_1^* & G_0 \end{bmatrix} \begin{bmatrix} U_s \\ U_i \end{bmatrix}$$

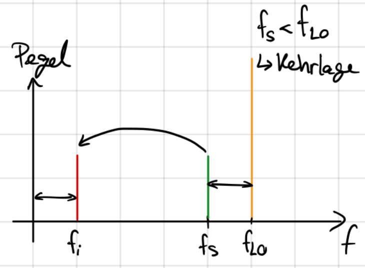
$\omega_s < \omega_p$ Kehrlage

$$\begin{bmatrix} I_s \\ I_i^* \end{bmatrix} = \begin{bmatrix} G_0 & G_1 \\ G_1^* & G_0 \end{bmatrix} \begin{bmatrix} U_s \\ U_i^* \end{bmatrix}$$

Frequenzumsetzer / Mischer

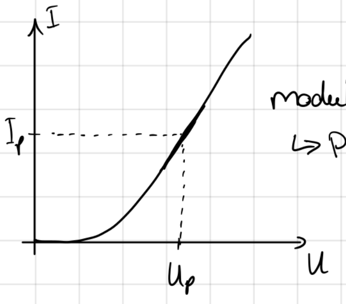






Abwärtsmischung

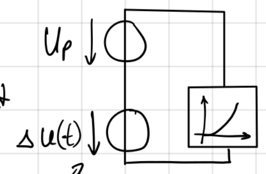
Für Umsetzung wird nichtlineares Bauelement benötigt. z.B. Schottky-Diode



moduliert AP $\sim \omega_p$

↳ Pumpsignal legt AP fest

Großsignal



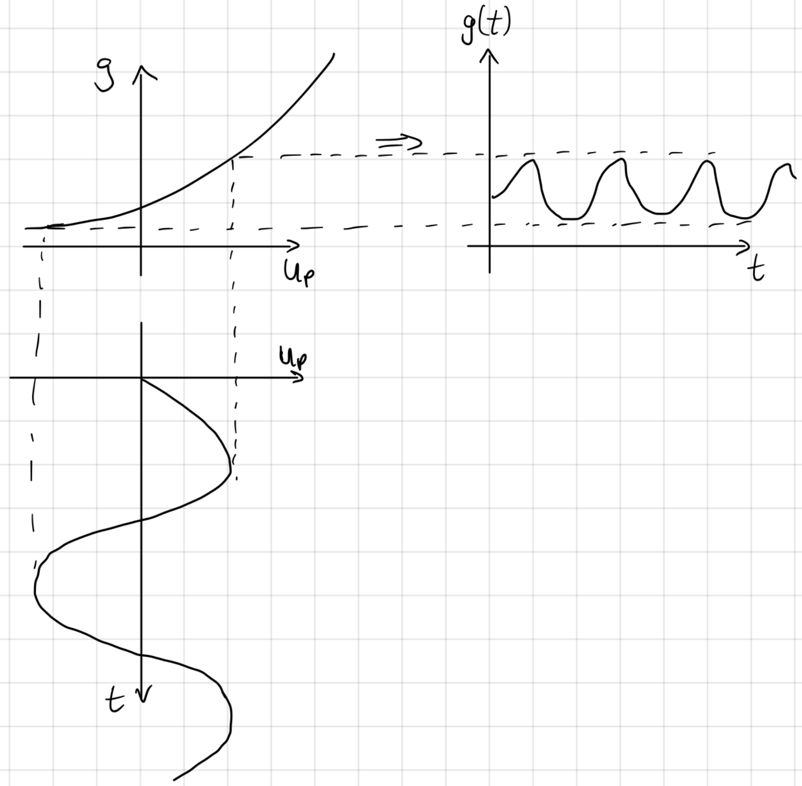
Kleinsignale, die den AP so vorfinden

$$I(u_p(t) + \Delta u_p(t)) = I(u_p(t)) + \underbrace{\underbrace{\left.\frac{dI}{dt}\right|_{u_p(t)}}_{g(u_p(t))} \cdot \Delta u(t)}_{\Delta i(t) = g(u_p(t)) \cdot \Delta u(t)}$$

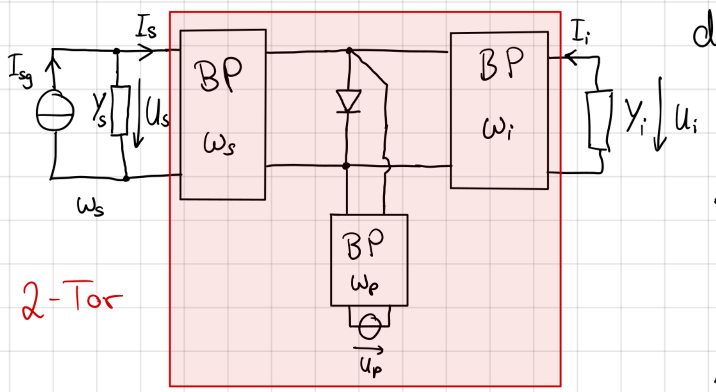
$$g(u_p(t)) = \sum_{n=-\infty}^{\infty} G_n e^{j\omega_p t}$$

$$G_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(u_p(t)) e^{-j\omega_p n t} \, d(\omega_p t)$$

$g(t)$ reell $\rightarrow G_{-n} = G_n^*$ G_0 reell



Abwärtsmischer



durch äußere Beschaltung mit BP-Filtern

=> nur Frequenzkomponenten bei ω_s, ω_p und ω_i

- Ansatz: es wird ein monofrequenter Kleinsignal mit ω_s angelegt, das den AP $\sim \omega_p$ vorfindet und nicht verändert

$$\Delta i(t) = g(u_p(t)) \cdot \Delta u(t) = \sum_{n=-\infty}^{\infty} G_n e^{jn\omega_p t} \cdot \Delta u(t)$$

$$\Delta i(t) = \frac{1}{2}\left[I_s e^{j\omega_s t} + I_s^* e^{-j\omega_s t} + I_i e^{j\omega_i t} + I_i^* e^{-j\omega_i t}\right]$$

$$= \sum_{n=-\infty}^{\infty} G_n e^{jn\omega_p t} \cdot \frac{1}{2}\left[U_s e^{j\omega_s t} + U_s^* e^{-j\omega_s t} + U_i e^{j\omega_i t} + U_i^* e^{-j\omega_i t}\right]$$

$$= \frac{1}{2}\Big[G_0 U_s e^{j\omega_s t} + G_0 U_s^* e^{-j\omega_s t} + G_0 U_i e^{j\omega_i t} + G_0 U_i^* e^{-j\omega_i t} + G_1 U_s^* e^{-j\omega_i t}$$

$$+ \underbrace{G_{-1}}_{G_1^*} U_s e^{j\overbrace{(-\omega_p+\omega_s)}^{}_{\omega_i} t} + G_1 U_i e^{j\underbrace{(\omega_p+\omega_i)}_{\omega_s} t} + G_{-1} U_i^* e^{j\underbrace{(-\omega_p-\omega_i)}_{-\omega_s} t}\Big]$$

für $n=1 \Rightarrow G_1 e^{j\omega_p t} \cdot \frac{1}{2} U_s e^{j\omega_s t}$

$$+ \underbrace{G_1 e^{j\omega_p t} \cdot \frac{1}{2} U_s^* e^{-j\omega_s t}}_{G_1 U_s^* e^{j\underbrace{(\omega_p-\omega_s)}_{-\omega_i} t}}$$

Gleichlage ($\omega_s > \omega_p$)

$\omega_i = \omega_s - \omega_p$, $\omega_s = \omega_i + \omega_p$

$-\omega_i = -\omega_s + \omega_p$, $-\omega_s = -\omega_i - \omega_p$

durch äußere Beschaltung nur Komponenten bei $\omega_i = \omega_s - \omega_p$ da Gleichlage mit $\omega_s > \omega_p$

nach Frequenzkomponenten:

$$\cancel{\tfrac{1}{2}} I_s \cancel{e^{j\omega_s t}} = \cancel{\tfrac{1}{2}}\left[G_0 U_s \cancel{e^{j\omega_s t}} + G_1 U_i \cancel{e^{j\omega_s t}}\right]$$

$$\Rightarrow I_s = G_0 U_s + G_1 U_i$$

$$\cancel{\tfrac{1}{2}} I_i \cancel{e^{j\omega_i t}} = \cancel{\tfrac{1}{2}}\left[G_0 U_i \cancel{e^{j\omega_i t}} + G_1^* U_s \cancel{e^{j\omega_i t}}\right]$$

$$I_i = G_0 U_i + G_1^* U_s$$

$$\begin{bmatrix} I_s \\ I_i \end{bmatrix} = \begin{bmatrix} G_0 & G_1 \\ G_1^* & G_0 \end{bmatrix} \begin{bmatrix} U_s \\ U_i \end{bmatrix}$$

Gleichlage mit $\omega_s > \omega_p$